



RUTGERS
NEW BRUNSWICK

AI for Visual Computing · CS 549

Dispatch - Autonomous Traffic & Emergency Response Simulation

Group: Rapid Route

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Project Overview and Motivation

- **What it is:** a browser-based 3D simulation of a city where hundreds of vehicles generate, move, and interact realistically, then an emergency layer (ambulances, fire, police) routes through that same live traffic.
- **The gap we're targeting:** most traffic demos are either arcade-style with no real logic, or academic routing demos with no visual believability. We want both at once - a simulation you can watch and trust.
- **Main goal:** get vehicles to move and interact like real traffic, generated by real demand, following each other safely, resolving junctions without collisions, before layering emergency dispatch on top and measuring how it performs under realistic conditions.

Proposed Framework & Technical Architecture

- **Rendering:** Three.js (WebGL) — 3D city, buildings, orbit/pan/zoom camera, vehicle trails and beacons.
- **Demand model:** zone-based city map; building density drives trip generation, an origin–destination (OD) matrix distributes trips across zones with peak/off-peak variation.
- **Microsimulation:** car-following (speed governed by gap to the vehicle ahead) + junction right-of-way rules, so vehicles never overlap or desync.
- **Routing:** A* over the road graph, cost-weighted by expected travel time (not raw distance) recomputed live as congestion and closures change.
- **Stack:** JavaScript/TypeScript, client-side, no backend required for the MVP.

Non-Trivial Challenges

Discrete Graph vs. Continuous Space Impedance Mismatch:

- *The Problem:* A* operates on zero-dimensional nodes and 1D edges, while visual computing occurs in continuous 2D/3D space with rigid bounding boxes, turning radii, and deceleration constraints.
- *Why AI Agents Fail:* Coding assistants can quickly scaffold an A* graph or write a basic separation force, but they repeatedly generate agents that jitter, freeze, clip through corners, or produce "ghost car" desynchronizations when switching from graph edge traversal to continuous collision steering.

Multi-Agent Deadlock & Right-of-Way at High Density:

- *The Problem:* At unsignalized or congested 4-way junctions, pairwise local avoidance (like naive Boids or social forces) creates symmetric stalemates where all four converging agents come to a dead stop.
- *Why AI Agents Fail:* LLM generators typically solve avoidance by adding repulsive forces, which actively increases jitter and oscillation in confined spaces. Hand-crafting priority reservation states and non-holonomic vehicle turning arcs requires manual mathematical tuning and profiling.

Dynamic Priority Egress & Saccadic Rerouting:

- Smoothly transitioning ambient civilian vehicles into yielding states (pulling to curbs) when emergency beacons enter sensory range, without invalidating their current trajectory splines.

Non-Trivial Challenges

- Emergent junction behavior at scale. Deadlocks, gridlock, and collision edge-cases only appear once hundreds of agents interact — they're not visible in isolated unit tests, only in the live emergent system.
- Believability tuning, not correctness. Car-following and avoidance parameters can be "technically correct" and still look robotic or jittery. Getting traffic to look real is a perceptual, iterative judgment call, not a pass/fail check.
- Cost-based routing calibration. Balancing distance vs. congestion vs. time needs realistic reference scenarios to validate against — there's no single ground truth to code toward.
- Performance under load. Hundreds of concurrent agents doing local sensing + pathfinding is a real profiling problem, not a one-time implementation.

Live 3D Prototype & Q&A

Interactive Demo: [Dispatch 3D Prototype](#)([UPDATE LINK](#))

Demo Showcase:

- Real-time 3D camera controls (Orbit, Pan, Zoom).
- Layered urban scene with density variations and dynamic road closures.
- Baseline A* path planning with live rerouting upon edge invalidation.